

# Circles: The Ultimate Study Guide

## Class 9th Mathematics (CBSE Geometry)

*From Fundamental Symmetries to High-Level Theorems and Proofs*

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Hello! Welcome to our comprehensive **Class 9 Circles Master Session**. Circles are one of the most elegant and symmetric shapes in geometry, appearing everywhere from nature to engineering. In this session, we will break down the entire Class 9 circles chapter from the basic vocabulary up to high-level proofs and NCERT exemplar problems. By the end of this guide, every concept in this chapter will be perfectly clear to you.

Let's begin our journey by looking at where circles come from and how they behave!

## Contents

# 1. Foundations: Symmetries & Key Definitions

**What is a Circle?** In nature, circles form when raindrops fall on water, and they are reflected in the cross-sections of plant stems, the inflorescences of sunflowers, the full moon, and the sun during a solar eclipse. Mathematically, we define a circle based on this uniform, symmetrical roundness:

## Mathematical Definition of a Circle

A **circle** is the set (or **locus**) of all points in a two-dimensional plane that lie at a constant, equal distance from a fixed point on that same plane.

The fixed point is called the **centre** of the circle, and the constant distance is called its **radius**.

**Symmetries of a Circle** Circles are perfectly symmetrical shapes, which is why rotating wheels look identical at all times:

- **Rotational Symmetry:** A circle has complete rotational symmetry about its centre through *any* angle. No matter how much you spin it, its shape looks exactly the same.
- **Reflection Symmetry:** A circle has complete reflection symmetry. If you fold a paper circle so that the boundaries overlap, the resulting crease represents a line of reflection symmetry. All such creases pass through the centre and represent the circle's **diameters**.

**Key Elements of a Circle** Before we dive into proofs, we must master our vocabulary:

1. **Radius ( $r$ ):** The line segment (or distance) from the centre of the circle to any point on its boundary.
2. **Chord:** A line segment joining any two distinct points on the circle's boundary.
3. **Diameter ( $d$ ):** A chord that passes through the centre of the circle. It is the longest chord in the circle, and its length is equal to twice the radius ( $d = 2r$ ). There is no chord longer than the diameter.
4. **Arc:** A connected portion of the circle's boundary. Two points  $A$  and  $B$  on a circle divide it into two arcs:
  - **Minor Arc:** The smaller portion of the boundary (subtending an angle  $< 180^\circ$  at the centre).
  - **Major Arc:** The larger portion of the boundary (subtending an angle  $> 180^\circ$  at the centre).
5. **Segment:** The region enclosed between a chord and its corresponding arc.
  - **Minor Segment:** Enclosed by the chord and the minor arc.
  - **Major Segment:** Enclosed by the chord and the major arc.

## 2. How Many Circles? (The 1, 2, and 3-Point Rules)

An essential question in geometry is: *Given a set of points, how many circles can we draw through them?*

**The One-Point Rule** Through a single point  $A$  on a plane, you can draw **infinitely many circles** of any size.

**The Two-Point Rule** Through two points  $A$  and  $B$  on a plane, you can draw **infinitely many circles**.

- **The Center Rule:** The centre of every circle passing through  $A$  and  $B$  must be equidistant from both points ( $OA = OB$ ). Therefore, the centres of all these circles lie on the **perpendicular bisector** of the line segment  $AB$ .
- **Smallest Circle:** The smallest possible circle passing through  $A$  and  $B$  is the one where  $AB$  is the diameter. Its radius is exactly  $\frac{1}{2}AB$ , and its centre is the midpoint of  $AB$ .

**The Three-Point Rule** This is a critical theorem in the CBSE syllabus:

### Theorem: Circles Through Three Points

**Statement:** There is one and only one (unique) circle passing through three given non-collinear points.

**Why is this true?** If we have three points  $A, B$ , and  $C$  that are **non-collinear** (not in a straight line):

1. Any circle passing through  $A$  and  $B$  must have its centre on the perpendicular bisector of  $AB$ .
2. Any circle passing through  $B$  and  $C$  must have its centre on the perpendicular bisector of  $BC$ .
3. Since  $A, B$ , and  $C$  are non-collinear, these two perpendicular bisectors are not parallel and will intersect at a **single, unique point  $O$** .
4. This intersection point  $O$  is equidistant from  $A, B$ , and  $C$  ( $OA = OB = OC$ ). Using  $O$  as the centre and  $OA$  as the radius, we can draw exactly one circle that passes through all three points.

**Circumcircle and Circumcentre** The unique circle passing through the three vertices of a triangle  $\triangle ABC$  is called the **circumcircle** of the triangle, and its centre  $O$  is called the **circumcentre**. The position of the circumcentre depends on the type of triangle:

### What about Three Collinear Points?

**No circle can pass through three collinear points.** Why? Because if  $A, B$ , and  $C$  lie on a straight line, the perpendicular bisectors of  $AB$  and  $BC$  are parallel to each other. They will never intersect, meaning there is no point that is equidistant from all three points!

| Type of Triangle | Position of Circumcentre ( $O$ )  | Visual Context                                       |
|------------------|-----------------------------------|------------------------------------------------------|
| Acute-Angled     | Inside the triangle               | Balanced sides, perpendiculars meet inside.          |
| Obtuse-Angled    | Outside the triangle              | Lies opposite the largest obtuse angle.              |
| Right-Angled     | At the midpoint of the hypotenuse | Hypotenuse acts as the diameter ( $90^\circ$ angle). |

Table 1: Location of the Circumcentre of a Triangle

### 3. CBSE Class 9 Circles Theorem Matrix

Here is a summary matrix of all core theorems for quick reference and revision:

## 4. Deep-Dive Theorem Proofs & Key Definitions

Let us master the formal geometric proofs of the most critical theorems. We will follow our exact logical structure for each.

### Theorem T1: Central Angles of Equal Chords

#### Core Intuition (WHY it works)

Think of a circle as a highly symmetrical wheel. Spinning the wheel doesn't deform it. If you have two chords of the exact same length, they form congruent triangular slices when connected to the center, meaning the angles at the center of those slices must be equal.

#### Theorem T1 Layout

**Statement:** Equal chords of a circle subtend equal angles at the centre.

**Visual Diagram Description:** Imagine a circle with its center labeled  $C$ . Along the bottom-left boundary, we have a chord  $AB$ , and along the bottom-right boundary, we have an identical chord  $DE$ . We draw the radii connecting the center  $C$  to the endpoints of both chords:  $CA$ ,  $CB$ ,  $CD$ , and  $CE$ . This creates two triangles,  $\triangle CAB$  and  $\triangle CDE$ , sharing a common vertex at the center  $C$ .

**Given:** A circle with centre  $C$  and two equal chords  $AB$  and  $DE$  (so  $AB = DE$ ).

**To Prove:**  $\angle ACB = \angle DCE$  (the central angles are equal).

**Step-by-Step Proof: Hence Proved!**  $\angle ACB = \angle DCE$ .

#### Key Revision Takeaway

Equal chords always cut out identical triangular wedges, resulting in equal angles at the center.

**Theorem T3 & T4: Perpendicular from Center to Chord** Let's explore the connection between the center, the perpendicular line, and the chord's midpoint.

#### Core Intuition (WHY it works)

A triangle formed by the center and any chord is always isosceles (since the other two sides are equal radii). In any isosceles triangle, the perpendicular dropped from the vertex to the base always bisects the base.

**Theorem T3 & T4 Layout**

**Statement (T3):** The perpendicular from the centre of a circle to a chord bisects the chord.

**Statement (T4 - Converse):** The line joining the centre of a circle and the midpoint of a chord is perpendicular to the chord.

**Visual Diagram Description:** Imagine a circle with center  $O$  and horizontal chord  $AB$ . Let  $M$  be the midpoint of  $AB$ . We draw a line segment from  $O$  to  $M$ . We also draw the radii  $OA$  and  $OB$ , creating two right-angled triangles  $\triangle OMA$  and  $\triangle OMB$  sharing the side  $OM$ .

**Step-by-Step Proof of T3 (Perpendicular  $\implies$  Bisector):**

- **Given:** A circle with centre  $O$ , chord  $AB$ , and line segment  $OM \perp AB$  (so  $\angle OMA = \angle OMB = 90^\circ$ ).
- **To Prove:**  $AM = BM$  (the perpendicular bisects the chord).

**Hence Proved!** The perpendicular from the centre bisects the chord ( $AM = BM$ ).

**Theorem T8: The Double Angle Theorem (Central vs. Circumference Angle)**

This is one of the most famous and highly tested circle theorems in CBSE examinations.

**Core Intuition (WHY it works)**

If you move the vertex of an angle from the circle's boundary to the center, the angle "opens up" and doubles in size because you are moving closer to the source of the arc.

**Theorem T8 Layout**

**Statement:** The angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.

**Visual Diagram Description:** Imagine a circle with centre  $C$  and an arc  $AFB$ . The angle subtended by arc  $AFB$  at the centre is  $\angle ACB$ . Let  $D$  be a point on the remaining part of the circle (outside arc  $AFB$ ). We connect  $D$  to  $C$  and extend the line  $DC$  to a point  $E$  on the circle. We also draw lines  $DA$  and  $DB$ . This creates two triangles,  $\triangle DCA$  and  $\triangle DCB$ .

**Given:** An arc  $AFB$  subtending angle  $\angle ACB$  at the centre  $C$ , and angle  $\angle ADB$  at point  $D$  on the boundary.

**To Prove:**  $\angle ACB = 2\angle ADB$ .

**Step-by-Step Proof: Hence Proved!** The angle at the centre is double the angle at the circumference.

**Key Revision Takeaway**

The central angle of any arc is always twice the angle that same arc makes anywhere on the circle's boundary.

## 5. Systematically Solved Practice Problems

To truly clear this chapter, let us apply our theorems to solve real-world and NCERT textbook problems step-by-step.

### Problem 1: Calculating Chord Length (Pythagorean Application)

#### Problem Statement

In a circle of radius 13 cm, a chord is drawn at a distance of 5 cm from the centre. Find the length of this chord.

**1. Identify Core Concept:** This problem uses **Theorem T3** (The perpendicular from the centre to a chord bisects the chord) and the **Baudhāyana-Pythagoras Theorem** ( $a^2 + b^2 = c^2$ ).

#### 2. Step-by-Step Solution:

1. Let the circle have its centre at  $O$  and radius  $R = 13$  cm. Let the chord be represented by the segment  $AB$ .
2. The distance from the centre to the chord is the perpendicular line segment  $OM$  dropped from  $O$  to  $AB$ . Thus,  $OM = 5$  cm and  $OM \perp AB$ .
3. Connect the centre  $O$  to endpoint  $A$ . The line segment  $OA$  is a radius of the circle, so  $OA = 13$  cm.
4. This forms a right-angled triangle  $\triangle OMA$  with the right angle at  $M$  ( $\angle OMA = 90^\circ$ ).
5. Apply the Pythagoras theorem to  $\triangle OMA$ :

$$OA^2 = OM^2 + AM^2$$

$$13^2 = 5^2 + AM^2$$

$$169 = 25 + AM^2$$

$$AM^2 = 169 - 25 = 144$$

$$AM = \sqrt{144} = 12 \text{ cm}$$

6. By Theorem T3, the perpendicular from the centre bisects the chord, which means  $M$  is the midpoint of  $AB$ , so:

$$AB = 2 \times AM = 2 \times 12 = 24 \text{ cm}$$

**Answer:** The length of the chord is **24 cm**.

#### Common Pitfall: The Half-Chord Trap

Many students solve for  $AM = 12$  cm and write that as their final answer. Remember,  $AM$  is only **half** of the chord length! Always read the question carefully and multiply by 2 to get the full length of chord  $AB$ .

## Problem 2: Finding a Nested Segment Length

### Problem Statement

In a circle of radius 5 cm, a chord  $AB$  of length 8 cm is drawn. A line segment  $OD$  is drawn from the centre  $O$  perpendicular to  $AB$ , intersecting  $AB$  at  $C$  and the circle boundary at  $D$ . Find the length of  $CD$ .

**1. Identify Core Concept:** This problem applies **Theorem T3** to split the chord, uses Pythagoras to find the perpendicular distance from the centre, and then subtracts this from the total radius to find the remaining segment.

### 2. Step-by-Step Solution:

1. Given the radius  $OA = 5$  cm and the chord  $AB = 8$  cm.
2. Since  $OD$  is perpendicular to  $AB$ , the perpendicular segment  $OC \perp AB$ .
3. By Theorem T3,  $OC$  bisects  $AB$ . Therefore,  $AC = BC = \frac{1}{2}AB = 4$  cm.
4. In right-angled triangle  $\triangle OCA$ :

$$OA^2 = OC^2 + AC^2$$

$$5^2 = OC^2 + 4^2$$

$$25 = OC^2 + 16$$

$$OC^2 = 25 - 16 = 9$$

$$OC = \sqrt{9} = 3 \text{ cm}$$

5. We know that  $OD$  is a line segment from the centre  $O$  to the boundary  $D$ , which means  $OD$  is a radius of the circle. Thus,  $OD = 5$  cm.
6. To find the length of segment  $CD$ :

$$CD = OD - OC = 5 - 3 = 2 \text{ cm}$$

**Answer:** The length of the segment  $CD$  is **2 cm**.

### Common Pitfall: Assuming $CD$ is half of something

Do not assume that the segment  $CD$  is half of the radius or equal to some arbitrary value. Always compute the perpendicular distance from the centre first, and then subtract it from the circle's known radius.

## 6. Quick-Recall Study Tips for CBSE Exams

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### Secret Key to Circles Chapter

Nearly 80% of all circle theorem proofs and numerical problems are solved using one simple trick: **Congruent Triangles** ( $\Delta \cong \Delta$ ) or the **Pythagoras Theorem** ( $a^2 + b^2 = c^2$ ).

### Exam Cheat-Sheet Checklist

- ✓ **Radii are always equal:** Whenever you see a circle problem with lines drawn to the boundary from the center, immediately label them as equal ( $r$ ). It instantly reveals hidden isosceles triangles!
- ✓ **The Semicircle Rule:** Any angle sitting on the diameter is  $90^\circ$ . If you see a diameter, look for a right-angled triangle.
- ✓ **Opposite Angles in Cyclic Quadrilaterals:** If you are given three corners of a cyclic quadrilateral, the fourth is just  $180^\circ$  minus the opposite corner.
- ✓ **The Center Perpendicular:** Whenever a line is perpendicular from the center to a chord, draw a radius to the chord's endpoint to form a right-angled triangle. This is the key to almost all chord distance problems.

Congratulations! You have successfully completed the core study session for the Class 9 Circles chapter. Keep this guide handy for reference and revision. Happy learning!

| #          | CBSE Core Theorem Statement                                                                      | Core Intuition (In Plain Words)                                                                  |
|------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>T1</b>  | Equal chords of a circle subtend equal angles at the centre.                                     | Identical chords make identical triangular wedges at the center.                                 |
| <b>T2</b>  | If the angles subtended by chords at the centre are equal, then the chords are equal.            | If the central "wedges" have the same angle, the chord lengths must match.                       |
| <b>T3</b>  | The perpendicular from the centre of a circle to a chord bisects the chord.                      | A straight vertical line dropped from the center splits a chord in half.                         |
| <b>T4</b>  | The line joining the centre and the mid-point of a chord is perpendicular to the chord.          | Connecting the center to a chord's exact middle always creates a $90^\circ$ angle.               |
| <b>T5</b>  | Equal chords of a circle are equidistant from the centre.                                        | Equal chords sit at the exact same depth from the circle's center.                               |
| <b>T6</b>  | Chords equidistant from the centre of a circle are equal in length.                              | Chords at equal depths from the center are of equal length.                                      |
| <b>T7</b>  | Given two unequal chords, the longer chord is closer to the centre.                              | The longer a chord is, the deeper it goes into the circle (closer to the center).                |
| <b>T8</b>  | The angle subtended by an arc at the centre is double the angle subtended by it at the boundary. | The "opening" angle at the center is always twice the angle made at the circle's edge.           |
| <b>T9</b>  | Angles in the same segment of a circle are equal.                                                | Any angles drawn from the same endpoints to the same segment are identical.                      |
| <b>T10</b> | The angle subtended by a diameter at any point on the circle is $90^\circ$ .                     | Any triangle built on the diameter has a perfect $90^\circ$ right angle at the boundary.         |
| <b>T11</b> | The sum of either pair of opposite angles of a cyclic quadrilateral is $180^\circ$ .             | Opposite corners of a four-sided shape in a circle always add up to $180^\circ$ .                |
| <b>T12</b> | If opposite angles of a quadrilateral sum to $180^\circ$ , the quadrilateral is cyclic.          | If opposite corners add to $180^\circ$ , you can draw a perfect circle through all four corners. |

Table 2: Summary Matrix of Class 9 Circles Theorems

| Step | Mathematical Statement                             | Geometric Reason                                             |
|------|----------------------------------------------------|--------------------------------------------------------------|
| 1.   | In triangles $\triangle CAB$ and $\triangle CDE$ : | We compare the corresponding sides of both triangles.        |
| 2.   | $CA = CD$                                          | Radii of the same circle ( $r$ ).                            |
| 3.   | $CB = CE$                                          | Radii of the same circle ( $r$ ).                            |
| 4.   | $AB = DE$                                          | Given (the chords are of equal length).                      |
| 5.   | $\triangle CAB \cong \triangle CDE$                | By <b>SSS (Side-Side-Side) Congruence Criterion</b> .        |
| 6.   | $\angle ACB = \angle DCE$                          | By <b>CPCT</b> (Corresponding Parts of Congruent Triangles). |

| Step | Mathematical Statement                                          | Geometric Reason                                      |
|------|-----------------------------------------------------------------|-------------------------------------------------------|
| 1.   | In right-angled triangles $\triangle OMA$ and $\triangle OMB$ : | We compare the two triangles on either side of $OM$ . |
| 2.   | $\angle OMA = \angle OMB = 90^\circ$                            | Given (since $OM \perp AB$ ).                         |
| 3.   | $OA = OB$                                                       | Radii of the same circle (Hypotenuse, $r$ ).          |
| 4.   | $OM = OM$                                                       | Common side (Side).                                   |
| 5.   | $\triangle OMA \cong \triangle OMB$                             | By <b>RHS (Right-Hypotenuse-Side) Congruence</b> .    |
| 6.   | $AM = BM$                                                       | By <b>CPCT</b> .                                      |

| Step | Mathematical Statement                                                | Geometric Reason                                          |
|------|-----------------------------------------------------------------------|-----------------------------------------------------------|
| 1.   | Join $D$ to centre $C$ and extend it to a point $E$ on the arc.       | Construction to split our angles.                         |
| 2.   | In $\triangle DCB$ , $CB = CD$                                        | Both are equal to the radius ( $r$ ).                     |
| 3.   | $\angle CBD = \angle CDB$                                             | Angles opposite to equal sides of a triangle.             |
| 4.   | $\angle BCE = \angle CBD + \angle CDB$                                | Exterior Angle Theorem ( $\angle BCE$ is exterior angle). |
| 5.   | $\angle BCE = 2\angle CDB = 2\angle BDC$                              | Substituting Step 3 into Step 4.                          |
| 6.   | Similarly, in $\triangle DCA$ : $\angle ACE = 2\angle CDA$            | Using the same logic for the left-hand triangle.          |
| 7.   | $\angle ACB = \angle BCE + \angle ACE$                                | Angle Addition.                                           |
| 8.   | $\angle ACB = 2\angle BDC + 2\angle CDA = 2(\angle BDC + \angle CDA)$ | Substituting Steps 5 and 6.                               |
| 9.   | $\angle ACB = 2\angle ADB$                                            | Since $\angle ADB = \angle BDC + \angle CDA$ .            |